

M^5 Midterm

Name: _____

1. State whether these statements are True or False. If False, give one example showing why. All variables represent real numbers.

a. $\sqrt{(x+1)^2} = x+1$

b. $(\sqrt{x} + 2\sqrt{y})^2 = x + 4y$

2. a. Give an example of a rational number between 3 and 4.

b. Give an example of an irrational number between 3 and 4.

3. Plot the set of real numbers x that satisfy $|x + 1| \geq 2$ and $x < 0$ on the number line.

4. Plot the set of real numbers x that satisfy $3(x - 1)^2 x(x + 3) < 0$ on the number line.

5. Find an equation of a line passing through the points $(-1, 3)$ and $(4, 7)$.

6. Simplify and eliminate any negative exponents (assume all variables are positive):

a. $\left(\frac{x^{1/2}}{y^{3/2}}\right)\left(\frac{y^{-1/2}}{x^3}\right).$

b. $\sqrt{18x^2y^6z^{-2}}$

7. Let $f(x) = x^2 - 4$.

a. Find $f(-2)$

b. Find $f(s/3)$.

c. Find all real numbers x that satisfy $f(x + 1) = 0$.

8. Sketch the graph of $y = \frac{x(x-3)}{(x-1)}$.

